

Project Proposal

1. Dataset

This project uses the Global Financial Development Database (GFDD), maintained by the World Bank. The GFDD is a widely cited, publicly available dataset that documents financial system characteristics for over 200 economies, with annual data from 1960 to 2021. It draws on records from the IMF's Financial Access Survey, the Bank for International Settlements, Bankscope, and other international sources.

The database is organised around a “4×2” framework, capturing four dimensions of financial systems—depth, access, efficiency, and stability—each measured separately for financial institutions (e.g. commercial banks, insurance companies) and financial markets (e.g. stock and bond markets). Together with structural and other indicators, the full dataset contains over 100 variables.

We can download the dataset through the following link:

<https://www.worldbank.org/en/publication/gfdr/data/global-financial-development-database>

Data Scope

I will work with a cross-sectional slice from 2019—the latest pre-COVID year with strong data coverage. After restricting to countries that have complete observations across all 23 selected indicators, the working dataset contains 111 countries and 23 continuous variables. The sample spans all four World Bank income groups (High: 31, Upper-Middle: 35, Lower-Middle: 34, Low: 11) and all seven geographic regions, giving a broad cross-section of global financial systems. One limitation of the working sample is the unequal group size across income categories (High: 31, Upper-Middle: 35, Lower-Middle: 34, Low: 11). The Low income group in particular is small relative to the others; this will be addressed in the LDA stage by specifying prior probabilities proportional to observed group frequencies rather than assuming equal priors.

The selected indicators are summarised in the table below:

Dimension	Code	Description	Scale
Depth	di01	Private credit by deposit money banks / GDP (%)	Continuous
Depth	di02	Deposit money banks' assets / GDP (%)	Continuous
Depth	di05	Liquid liabilities / GDP (%)	Continuous
Depth	di08	Financial system deposits / GDP (%)	Continuous
Depth	di14	Domestic credit to private sector / GDP (%)	Continuous
Access	ai02	Bank branches per 100,000 adults	Continuous
Access	ai25	ATMs per 100,000 adults	Continuous
Efficiency	ei01	Bank net interest margin (%)	Continuous
Efficiency	ei03	Bank noninterest income to total income (%)	Continuous
Efficiency	ei04	Bank overhead costs / total assets (%)	Continuous
Efficiency	ei05	Bank return on assets (% , after tax)	Continuous
Efficiency	ei06	Bank return on equity (% , after tax)	Continuous
Efficiency	ei07	Bank cost-to-income ratio (%)	Continuous

Efficiency	ei08	Credit to government and SOEs / GDP (%)	Continuous
Efficiency	ei09	Bank return on assets (% , before tax)	Continuous
Efficiency	ei10	Bank return on equity (% , before tax)	Continuous
Stability	si01	Bank Z-score	Continuous
Stability	si04	Bank credit to bank deposits (%)	Continuous
Stability	si06	Liquid assets to deposits and short-term funding (%)	Continuous
Other	oi01	Bank concentration (%)	Continuous
Other	oi02	Bank deposits / GDP (%)	Continuous
Other	oi13	Remittance inflows / GDP (%)	Continuous
Other	oi14	Consolidated foreign claims of BIS reporting banks / GDP (%)	Continuous

Categorical grouping variable: income (World Bank income classification with 4 levels: High, Upper-Middle, Lower-Middle, and Low income). An alternative grouping variable, region (7 levels), is also available for sensitivity analyses.

2. Research Question

The central question is: *how well do financial development indicators characterise and differentiate economies at different income levels, and what latent structure underlies cross-country variation in financial systems?*

Wealthier countries generally have deeper and more efficient banking sectors, but which specific dimensions of financial development—depth, physical access, operational efficiency, or systemic stability—are most informative for distinguishing income groups is not obvious from the literature alone. The 23 indicators also show considerable overlap: several depth variables (private credit, bank assets, liquid liabilities) measure closely related aspects of the same underlying construct, and the before-tax and after-tax profitability ratios are nearly collinear. These correlations make dimensionality reduction a sensible first step before any classification analysis.

A secondary question is whether countries form coherent clusters in the reduced-dimension financial space that correspond to known economic groupings, or whether some economies are more heterogeneous than their income label suggests.

3. Methodological Approach

Because the 23 variables are measured on quite different scales (e.g. the Z-score ranges from about 1 to 55, while liquid liabilities can exceed 600% of GDP), all analyses will be carried out on standardised data. The planned analysis has several stages:

Stage 1: Principal Component Analysis (PCA)

PCA will be the starting point for exploring the covariance structure of the 23 indicators and reducing dimensionality. I will examine eigenvalue decay via a scree plot, the cumulative variance explained, and biplot representations to see which variables group together. Of particular interest is whether the resulting components correspond to the four theoretical dimensions (depth, access, efficiency, stability), or whether the

empirical structure cuts across these categories.

Stage 2: Linear Discriminant Analysis (LDA)

LDA will be applied to the retained PCA scores—rather than the raw indicators—to find the linear combinations that best separate the four income groups. This approach avoids the numerical instability introduced by near-collinear variables such as the before- and after-tax profitability pairs (ei05/ei09, ei06/ei10), while retaining the majority of the variance in the original data. This will produce discriminant functions that can be visualised in a low-dimensional scatter, together with a confusion matrix and cross-validated classification accuracy. If the multivariate normality or equal-covariance assumptions of LDA appear problematic, I may also consider quadratic discriminant analysis as an alternative.

Stage 3: Cluster Analysis

As a complement to the supervised classification from LDA, I will run a cluster analysis on the standardised indicators—most likely hierarchical clustering with Ward's method, possibly followed by k-means refinement. The main question here is whether the unsupervised groupings align with the World Bank income categories, or whether some countries sit closer to a different group once financial development is accounted for. Silhouette widths will be used to guide the choice of cluster number.

Stage 4: Multidimensional Scaling (MDS)

Time permitting, I may also apply classical or non-metric MDS to produce a two-dimensional map of the 111 countries from their pairwise distances in the indicator space. This would give a visualisation that does not impose linear structure, and should make outlier economies—financial centres like Luxembourg or Hong Kong, for instance—stand out clearly relative to their income-group peers.

Potential Extensions

Should time and scope allow, further extensions could include Canonical Correlation Analysis to examine the relationship between the depth/access block and the efficiency/stability block of variables, MANOVA to formally test for mean differences across income groups on the multivariate response, or Partial Least Squares (PLS/PLS-DA) as a collinearity-robust alternative to LDA.